

		Pr > F
Source		0.0023
IQ		0.8288
GROUP		
Covariate Example		
The GLM Procedure		
Least Squares Means		
GROUP	MATH	LSMEAN
A		392.345889
B		400.654111

PROBLEMS

Remember, you can download all the data sets and programs for these problems from the web site: www.prenhall.com/cody

- 7.1 Problems 7.1 and 7.3 were inspired by one of the authors (Cody) watching the French Open Tennis tournament while working on problem sets. (McEnroe versus Lendl (1984). Lendl won in five sets.)

Three brands of tennis shoes are tested to see how many months of playing would wear out the soles. Eight pairs of brands A, N, and T are randomly assigned to a group of 24 volunteers. The following table shows the results of the study:

	Brand A	Brand N	Brand T
Wear time, in months	8	4	12
	10	7	8
	9	5	10
	11	5	10
	10	6	11
	10	7	9
	8	6	9
	12	4	12

Are the brands equal in wear quality? Write a SAS program to solve this problem, using ANOVA. Include the statements to perform an SNK multiple comparison.

- 7.2 Two cholesterol-lowering medications (statins) and a placebo were given to each of 10 volunteers with total cholesterol readings of 240 or higher. After 6 weeks, the following total cholesterol values were recorded:

Statin A: 220 190 180 185 210 170 178 200 177 189

Statin B: 160 168 178 200 172 155 159 167 185 199

Placebo: 240 220 246 244 198 238 277 255 190 188

Create a SAS data set by reading these data. As an interesting exercise, try reading three lines like this:

```
220 190 180 185 210 170 178 200 177 189
160 168 178 200 172 155 159 167 185 199
240 220 246 244 198 238 277 255 190 188
```

That is, have the program create the treatment variable (call it TREAT) with values of 'A', 'B', and 'Placebo'. Next run a oneway ANOVA followed by a Student-Newman-Keuls multiple-comparison test.

- 7.3 Tennis balls are tested in a machine to see how many bounces they can withstand before they fail to bounce 30% of their dropping height. Two brands of balls (W and P) are compared. In addition, the effect of shelf life on these brands is tested. Half of the balls of each brand are 6 months old, the other half, fresh. Using a two-way analysis of variance, what conclusions can you reach? The data are shown here:

		Brand W	Brand P
Age	New	67 72 74 82 81	75 76 80 72 73
	Old	46 44 45 51 43	63 62 66 62 60

- 7.4 Four different methods of preparing for a college entrance exam were compared. They are labeled 'A', 'B', 'C', and 'D'. The following exam scores were obtained for each of the four programs:

Program							
A	560	520	530	525	575	527	580
B	565	522	520	530	510	522	600
C	512	518	555	502	510	520	516
D	505	508	512	520	543	523	517

Compare the four preparation methods. Use a multiple-comparison method of your choice to make pairwise comparisons. Create two contrasts: One to compare methods A and B to C and D, the other to compare method D to the other three.

- 7.5 A taste test is conducted to rate consumer preference between brands C and P of a popular soft drink. In addition, the age of the respondent is recorded (1 = less than 20, 2 = 20 or more). Preference data (on a scale of 1–10) is as follows:

Age		Brand C	Brand P
<20		7	9
		6	8
		6	9
		5	9
		6	8
>=20		9	6
		8	7
		8	6
		9	6
		7	5
		8	
		8	

- (a) Write a SAS program to analyze these data with a two-way analysis of variance. (Careful: Is the design balanced?) Note: Go ahead and treat these data as interval data even though some analysts would prefer that you use a nonparametric analysis.
- (b) Use SAS to plot an interaction graph.
- (c) Follow up with a t-test comparing brand C to brand P for each age group separately.
- 7.6 Two groups of patients, those with a genetic deficiency and the other a "normal" group, are given either a drug to cure depression or a placebo. A standardized depression survey is given to all the subjects with the following results:

Genetic Deficiency		Anti-Depression Drug	Placebo
		9	9
		11	6
		10	6
		10	7
Normal		5	12
		4	11
		7	10
		7	11

- Perform a two-way analysis of variance to test for drug and genetic differences.
- 7.7 A manufacturer wants to reanalyze the data of problem 7.1, omitting all data for brand N. Run the appropriate analysis.

- 7.8 Run the following program to create the data set CO_VARY. Compute the correlation between IQ and SCORE and conduct a t-test comparing the two groups on IQ and SCORE. Next, test if the slope of IQ versus SCORE is homogeneous across groups. Finally, conduct an analysis of covariance, testing if the mean SCORE is different by GROUP after you have adjusted for IQ.

```
DATA CO_VARY;
  DO I = 1 TO 20;
    DO GROUP = 'A', 'B';
      SUBJ + 1;
      IQ = INT(RANNOR(124)*10 + 120 + 15*(GROUP EQ 'A'));
      SCORE = INT(.7*IQ + RANNOR(0)*10 + 100 + 10*(GROUP EQ 'B'));
      OUTPUT;
    END;
  END;
  DROP I;
RUN;
PROC PRINT DATA=CO_VARY NOOBS;
  TITLE "Listing of CO_VARY";
RUN;
```

- 7.9 What's wrong with this program?

```
1 DATA TREE;
2   INPUT TYPE $ LOCATION $ HEIGHT;
3 DATALINES;
4   PINE NORTH 35
5   PINE NORTH 37
6   PINE NORTH 41
7   PINE NORTH 41
8   MAPLE NORTH 44
9   MAPLE NORTH 41
10  PINE SOUTH 53
11  PINE SOUTH 55
12  MAPLE SOUTH 28
13  MAPLE SOUTH 33
14  MAPLE SOUTH 32
15  MAPLE SOUTH 22
16 ;
17 PROC ANOVA DATA=TREE;
18   CLASS TYPE LOCATION;
19   MODEL HEIGHT = TYPE | LOCATION;
20   MEANS TYPE LOCATION TYPE*LOCATION;
21 RUN;
```